

1995 年数学(三) 真题解析

一、填空题

(1) 【答案】 $\frac{2(-1)^n n!}{(x+1)^{n+1}}$.

【解】 由 $f(x) = \frac{1-x}{1+x} = -1 + \frac{2}{x+1}$ 得 $f^{(n)}(x) = \frac{2(-1)^n n!}{(x+1)^{n+1}}$.

(2) 【答案】 $2xyf\left(\frac{y}{x}\right)$.

【解】 $z'_x = yf - \frac{y^2}{x}f'$, $z'_y = xf + yf'$,

则 $xz'_x + yz'_y = 2xyf\left(\frac{y}{x}\right)$.

(3) 【答案】 $x + e^x + C$.

【解】 由 $f'(\ln x) = 1 + x$, 即 $f'(\ln x) = 1 + e^{\ln x}$ 得 $f'(x) = 1 + e^x$,
故 $f(x) = x + e^x + C$.

(4) 【答案】 $\frac{1}{10} \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 4 & 5 \end{pmatrix}$.

【解】 $|A| = 10$, 由 $A^* = |A|A^{-1} = 10A^{-1}$ 得

$$(A^*)^{-1} = \frac{1}{10}A = \frac{1}{10} \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 4 & 5 \end{pmatrix}.$$

(5) 【答案】 $\bar{X} \cdot \frac{\sqrt{n(n-1)}}{Q}$.

【解】 $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} Q^2$,

因为参数 μ 与 σ^2 未知, 所以对假设 $H_0: \mu = 0$, 应取的统计量为

$$t = \frac{\bar{X}}{\frac{S}{\sqrt{n}}} = \bar{X} \cdot \frac{\sqrt{n(n-1)}}{Q}.$$

二、选择题

(1) 【答案】 (D).

【解】 由 $-1 = \lim_{x \rightarrow 0} \frac{f(1) - f(1-x)}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{f(1-x) - f(1)}{-x} = \frac{1}{2} f'(1)$,

得 $f'(1) = -2$, 应选(D).

(2) 【答案】 (A).

【解】 方法一 对 $\int_0^1 \frac{1}{\sin x} dx$, 因为 $\lim_{x \rightarrow 0^+} (x-0)^1 \cdot \frac{1}{\sin x} = 1$ 且 $1 \geq 1$, 所以 $\int_0^1 \frac{1}{\sin x} dx$ 发散,

从而 $\int_{-1}^1 \frac{1}{\sin x} dx$ 发散, 应选(A).

方法二 因为 $\lim_{x \rightarrow -1^+} (x+1)^{\frac{1}{2}} \cdot \frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{2}}$, $\lim_{x \rightarrow 1^-} (1-x)^{\frac{1}{2}} \cdot \frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{2}}$ 且 $\frac{1}{2} < 1$,

所以 $\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx$ 收敛;

因为 $\int_0^{+\infty} e^{-x^2} dx \stackrel{x^2=t}{=} \frac{1}{2} \int_0^{+\infty} t^{-\frac{1}{2}} e^{-t} dt = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$, 所以 $\int_0^{+\infty} e^{-x^2} dx$ 收敛;

由 $\int_2^{+\infty} \frac{1}{x \ln^2 x} dx = -\frac{1}{\ln x} \Big|_2^{+\infty} = \frac{1}{\ln 2}$, 得 $\int_2^{+\infty} \frac{1}{x \ln^2 x} dx$ 收敛, 故应选(A).

(3) 【答案】 (C).

【解】 由 $BA = O$ 得 $r(B) + r(A) \leq m$, 再由 $r(A) = m$ 得 $r(B) = 0$, 即 $B = O$, 应选(C).

(4) 【答案】 (D).

【解】 因为 X, Y 独立同分布,

所以 $\text{Cov}(U, V) = \text{Cov}(X - Y, X + Y) = \text{Cov}(X, X) - \text{Cov}(Y, Y) = D(X) - D(Y) = 0$,

于是 $\rho_{UV} = \frac{\text{Cov}(U, V)}{\sqrt{D(U)} \cdot \sqrt{D(V)}} = 0$, 应选(D).

(5) 【答案】 (C).

【解】 由 $X \sim N(\mu, \sigma^2)$ 得 $\frac{X - \mu}{\sigma} \sim N(0, 1)$,

则 $P\{|X - \mu| < \sigma\} = P\left\{-1 < \frac{X - \mu}{\sigma} < 1\right\} = \Phi(1) - \Phi(-1)$,

即概率不变, 应选(C).

三、【解】 $f(0-0) = \lim_{x \rightarrow 0^-} \frac{2(1 - \cos x)}{x^2} = 1$, $f(0+0) = \lim_{x \rightarrow 0^+} \frac{\int_0^x \cos t^2 dt}{x} = \lim_{x \rightarrow 0^+} \cos x^2 = 1$,

因为 $f(0-0) = f(0) = f(0+0) = 1$, 所以 $f(x)$ 在 $x = 0$ 处连续.

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} &= \lim_{x \rightarrow 0^-} \frac{\frac{2(1 - \cos x)}{x^2} - 1}{x} = \lim_{x \rightarrow 0^-} \frac{2(1 - \cos x) - x^2}{x^3} \\ &= \lim_{x \rightarrow 0^-} \frac{2\sin x - 2x}{3x^2} = \frac{2}{3} \lim_{x \rightarrow 0^-} \frac{\sin x - x}{x^2} = 0, \end{aligned}$$

则 $f'_-(0) = 0$;

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\frac{\int_0^x \cos t^2 dt}{x} - 1}{x} = \lim_{x \rightarrow 0^+} \frac{\int_0^x \cos t^2 dt - x}{x^2} = \lim_{x \rightarrow 0^+} \frac{\cos x^2 - 1}{2x} = 0,$$

则 $f'_+(0) = 0$,

因为 $f'_-(0) = f'_+(0) = 0$, 所以 $f(x)$ 在 $x = 0$ 处可导, 且 $f'(0) = 0$.

四、【解】 由 $f(x) = \int_0^{3x} f\left(\frac{t}{3}\right) dt + e^{2x}$ 得

$$f'(x) - 3f(x) = 2e^{2x},$$

解得 $f(x) = \left(\int 2e^{2x} \cdot e^{-3dx} dx + C \right) e^{-\int -3dx} = (-2e^{-x} + C)e^{3x}$,

即 $f(x) = Ce^{3x} - 2e^{2x}$,

再由 $f(0) = 1$ 得 $C = 3$, 故 $f(x) = 3e^{3x} - 2e^{2x}$.

五、【解】 $y = \ln[(1+x)(1-2x)] = \ln(1+x) + \ln(1-2x)$,

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n, \text{ 其中 } -1 < x \leq 1;$$

$$\ln(1-2x) = -\sum_{n=1}^{\infty} \frac{2^n}{n} x^n, \text{ 其中 } -\frac{1}{2} \leq x < \frac{1}{2},$$

$$\text{故 } \ln(1-x-2x^2) = \sum_{n=1}^{\infty} \left[\frac{(-1)^{n-1}}{n} - \frac{2^n}{n} \right] x^n, \text{ 其收敛区间为 } \left(-\frac{1}{2}, \frac{1}{2}\right).$$

六、【解】 方法一 令 $D = \{(x, y) \mid x^2 + y^2 \leq R^2\}$, 则

$$\begin{aligned} & \iint_D \min\{x, y\} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{\frac{\pi}{4}} \sin \theta d\theta \int_0^R r^2 e^{-r^2} dr + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \cos \theta d\theta \int_0^R r^2 e^{-r^2} dr + \int_{\frac{5\pi}{4}}^{2\pi} \sin \theta d\theta \int_0^R r^2 e^{-r^2} dr \\ &= \left(1 - \frac{\sqrt{2}}{2}\right) \int_0^R r^2 e^{-r^2} dr - \sqrt{2} \int_0^R r^2 e^{-r^2} dr - \left(1 + \frac{\sqrt{2}}{2}\right) \int_0^R r^2 e^{-r^2} dr \\ &= -2\sqrt{2} \int_0^R r^2 e^{-r^2} dr, \end{aligned}$$

$$\begin{aligned} \text{故 } \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \min\{x, y\} e^{-(x^2+y^2)} dx dy &= \lim_{R \rightarrow +\infty} \iint_D \min\{x, y\} e^{-(x^2+y^2)} dx dy \\ &= -2\sqrt{2} \int_0^{+\infty} r^2 e^{-r^2} dr \stackrel{r^2=t}{=} -\sqrt{2} \int_0^{+\infty} t^{\frac{1}{2}} e^{-t} dt \\ &= -\sqrt{2} \Gamma\left(\frac{1}{2} + 1\right) = -\frac{\sqrt{2}\pi}{2}. \end{aligned}$$

方法二 令 $D_1 = \{(x, y) \mid y < x\}$, 由对称性得

$$\begin{aligned} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \min\{x, y\} e^{-(x^2+y^2)} dx dy &= 2 \iint_{D_1} y e^{-(x^2+y^2)} dx dy = 2 \int_{-\frac{3}{4}\pi}^{\frac{\pi}{4}} d\theta \int_0^{+\infty} r^2 \sin \theta \cdot e^{-r^2} dr \\ &= 2 \int_{-\frac{3}{4}\pi}^{\frac{\pi}{4}} \sin \theta d\theta \int_0^{+\infty} r^2 e^{-r^2} dr = -2\sqrt{2} \int_0^{+\infty} r^2 e^{-r^2} dr \\ &\stackrel{r^2=t}{=} -\sqrt{2} \int_0^{+\infty} t^{\frac{1}{2}} e^{-t} dt = -\sqrt{2} \Gamma\left(\frac{1}{2} + 1\right) = -\frac{\sqrt{2}\pi}{2}. \end{aligned}$$

七、【解】 $\frac{dR}{dQ} = P + Q \frac{dP}{dQ} = P + P \cdot \frac{dP/dQ}{P/Q} = \left(1 - \frac{1}{b}\right)P,$

由 $\left. \frac{dR}{dQ} \right|_{Q=Q_0} = a$, 得 $P_0 = \frac{ab}{b-1}$;

$$\frac{dR}{dP} = Q + P \frac{dQ}{dP} = Q + Q \cdot \frac{dQ/dP}{Q/P} = (1-b)Q,$$

由 $\left. \frac{dR}{dP} \right|_{P=P_0} = c < 0$ 得 $Q_0 = \frac{c}{1-b}$.

八、(1)【证明】 因为 $g(-x) = g(x)$,

$$\begin{aligned} \text{所以 } \int_{-a}^a f(x)g(x)dx &= \int_0^a [f(-x)g(-x) + f(x)g(x)]dx \\ &= \int_0^a [f(-x) + f(x)]g(x)dx = A \int_0^a g(x)dx. \end{aligned}$$

$$(2) \text{【解】} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^x dx = \int_0^{\frac{\pi}{2}} (\arctan e^x + \arctan e^{-x}) \sin x dx,$$

$$\text{因为} (\arctan e^x + \arctan e^{-x})' = \frac{e^x}{1+e^{2x}} - \frac{e^{-x}}{1+e^{-2x}} = 0,$$

所以 $\arctan e^x + \arctan e^{-x} \equiv A$,

$$\text{取 } x=0 \text{ 得 } A = \frac{\pi}{2},$$

$$\text{故} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan e^x dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin x dx = \frac{\pi}{2}.$$

九、【证明】 方法一 因为 $r(I) = 3$, 所以向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关,

又因为 $r(II) = 3$, 所以向量 α_4 可由向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性表示,

不妨设 $\alpha_4 = l_1 \alpha_1 + l_2 \alpha_2 + l_3 \alpha_3$,

再由 $r(III) = 4$ 得向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_5$ 线性无关,

$$\text{令} \quad k_1 \alpha_1 + k_2 \alpha_2 + k_3 \alpha_3 + k_4 (\alpha_5 - \alpha_4) = \mathbf{0},$$

代入整理得

$$(k_1 - l_1 k_4) \alpha_1 + (k_2 - l_2 k_4) \alpha_2 + (k_3 - l_3 k_4) \alpha_3 + k_4 \alpha_5 = \mathbf{0},$$

因为 $\alpha_1, \alpha_2, \alpha_3, \alpha_5$ 线性无关, 所以 $k_1 - l_1 k_4 = k_2 - l_2 k_4 = k_3 - l_3 k_4 = k_4 = 0$,

从而 $k_1 = k_2 = k_3 = k_4 = 0$, 即向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_5 - \alpha_4$ 线性无关,

故向量组 $\alpha_1, \alpha_2, \alpha_3, \alpha_5 - \alpha_4$ 的秩为 4.

方法二 由向量组 (I) 与 (II) 的秩为 3 得 $\alpha_1, \alpha_2, \alpha_3$ 线性无关且 α_4 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示,

不妨设 $\alpha_4 = k_1 \alpha_1 + k_2 \alpha_2 + k_3 \alpha_3$;

由向量组 (III) 的秩为 4 得 $\alpha_1, \alpha_2, \alpha_3, \alpha_5$ 线性无关, 令 $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_5)$, 则 $r(A) = 4$,

令 $B = (\alpha_1, \alpha_2, \alpha_3, \alpha_5 - \alpha_4) = (\alpha_1, \alpha_2, \alpha_3, -k_1 \alpha_1 - k_2 \alpha_2 - k_3 \alpha_3 + \alpha_5)$, 则

$$B = A \begin{pmatrix} 1 & 0 & 0 & -k_1 \\ 0 & 1 & 0 & -k_2 \\ 0 & 0 & 1 & -k_3 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\text{因为} \begin{pmatrix} 1 & 0 & 0 & -k_1 \\ 0 & 1 & 0 & -k_2 \\ 0 & 0 & 1 & -k_3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{可逆, 所以 } r(B) = r(A) = 4, \text{ 从而 } \alpha_1, \alpha_2, \alpha_3, \alpha_5 - \alpha_4 \text{ 线性无关.}$$

$$\text{十、【解】} (1) \text{ 令 } A = \begin{pmatrix} 0 & 2 & -2 \\ 2 & 4 & 4 \\ -2 & 4 & -3 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix},$$

$$\text{则 } f(x_1, x_2, x_3) = X^T A X.$$

$$(2) \text{ 由 } |\lambda E - A| = \begin{vmatrix} \lambda & -2 & 2 \\ -2 & \lambda - 4 & -4 \\ 2 & -4 & \lambda + 3 \end{vmatrix} = (\lambda + 6)(\lambda - 1)(\lambda - 6) = 0, \text{ 得 } \lambda_1 = -6, \lambda_2 = 1, \lambda_3 = 6,$$

$$\text{由 } 6E + A = \begin{pmatrix} 6 & 2 & -2 \\ 2 & 10 & 4 \\ -2 & 4 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 5 & 2 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得}$$

$\lambda_1 = -6$ 对应的线性无关的特征向量为 $\alpha_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$;

由 $E - A = \begin{pmatrix} 1 & -2 & 2 \\ -2 & -3 & -4 \\ 2 & -4 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 得

$\lambda_2 = 1$ 对应的线性无关的特征向量为 $\alpha_2 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$;

由 $6E - A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 2 & -4 \\ 2 & -4 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{5}{2} \\ 0 & 0 & 0 \end{pmatrix}$, 得

$\lambda_3 = 6$ 对应的线性无关的特征向量为 $\alpha_3 = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$,

令 $\gamma_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, $\gamma_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$, $\gamma_3 = \frac{1}{\sqrt{30}} \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$,

所求的正交矩阵为 $Q = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ -\frac{1}{\sqrt{6}} & 0 & \frac{5}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \end{pmatrix}$.

作正交变换 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = Q \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, 则二次型 f 化为标准形 $f = -6y_1^2 + y_2^2 + 6y_3^2$.

十一、【解】 令 $A = \{\text{仪器需要进一步检测}\}$, $B = \{\text{仪器可以出厂}\}$,

$AB = \{\text{仪器经过检测后可以出厂}\}$, 则 $B = \bar{A} + AB$,

$$P(A) = 0.3, \quad P(B | A) = 0.8, \quad P(AB) = P(A)P(B | A) = 0.24,$$

由概率的有限可加性得 $P(B) = P(\bar{A}) + P(AB) = 0.94$.

设 X 表示 n 台仪器中能出厂的台数, 则 $X \sim B(n, 0.94)$, 故

$$(1) \alpha = P\{X = n\} = 0.94^n;$$

$$(2) \beta = C_n^{n-2} \cdot 0.94^{n-2} \cdot 0.06^2;$$

$$(3) \theta = P\{X \leq n-2\} = 1 - P\{X = n\} - P\{X = n-1\} = 1 - n \cdot 0.94^{n-1} \cdot 0.06 - 0.94^n.$$

十二、【解】 $F(x, y) = P\{X \leq x, Y \leq y\} = \int_{-\infty}^x dx \int_{-\infty}^y f(x, y) dy$.

当 $x < 0$ 或 $y < 0$ 时, $F(x, y) = 0$;

$$\text{当 } 0 \leq x \leq 1, 0 \leq y \leq 1 \text{ 时, } F(x, y) = \int_0^x 4u du \int_0^y v dv = x^2 y^2;$$

当 $0 \leq x \leq 1, y > 1$ 时, $F(x, y) = \int_0^x 4u \, du \int_0^1 v \, dv = x^2$;

当 $x > 1, 0 \leq y \leq 1$ 时, $F(x, y) = \int_0^1 4u \, du \int_0^y v \, dv = y^2$;

当 $x > 1, y > 1$ 时, $F(x, y) = \int_0^1 4u \, du \int_0^1 v \, dv = 1$;

$$\text{故 } F(x, y) = \begin{cases} x^2 y^2, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ x^2, & 0 \leq x \leq 1, y > 1, \\ y^2, & x > 1, 0 \leq y \leq 1, \\ 1, & x > 1, y > 1, \\ 0, & \text{其他.} \end{cases}$$